

Chapter 5: Work, Energy and Power

Key Concepts

Work (W) — Work is done when a force moves an object through a distance in the direction of the force. $W = F \times d$, where F is force parallel to displacement and d is displacement. SI unit: joule (J). $1 \text{ J} = 1 \text{ N m}$.

Positive Work — Work done when force and displacement are in the same direction (, $\theta = 0^\circ$) Example: pushing a box forward.

Negative Work — Work done when force and displacement are in opposite directions ($\theta = 180^\circ$). Example: friction opposing motion.

Zero Work — Work done when force is perpendicular to displacement ($\vec{F} \perp \vec{d}$).
Example: carrying a bag while walking horizontally (vertical force, horizontal displacement).

Energy (E) — The capacity of a body to do work. SI unit: joule (J). Energy is a scalar quantity.

Kinetic Energy (KE) — Energy possessed by a body due to its motion. $KE = \frac{1}{2}mv^2$, where m is mass and v is velocity.

Potential Energy (PE) — Energy possessed by a body due to its position or state. Gravitational PE = mgh , where h is height above a reference level.

Mechanical Energy — The sum of kinetic and potential energy in a system. $ME = KE + PE$.

Chemical Energy — Energy stored in chemical bonds of substances (food, batteries, fuels).

Electrical Energy — Energy due to the flow of electric charge.

Heat Energy (Thermal Energy) — Energy due to the random motion of particles in a substance.

Light Energy (Radiant Energy) — Energy carried by light waves (electromagnetic radiation).

Nuclear Energy — Energy released during nuclear reactions (fission, fusion).

Sound Energy — Energy carried by sound waves (vibrations of particles).

Law of Conservation of Energy — Energy cannot be created nor destroyed. It can only be transformed from one form to another. Total energy in an isolated system remains constant.

Power (P) — The rate at which work is done or energy is transferred. $P = W / t$, where W is work and t is time. SI unit: watt (W). $1 \text{ W} = 1 \text{ J/s}$. Also: $P = F \times v$.

Efficiency (η)— The ratio of useful output energy (or power) to total input energy (or power). $\eta = \frac{\text{Useful Output Energy (or Power)}}{\text{Total Input Energy (or Power)}} \times 100\%$. Always less than 100% due to energy losses.

Renewable Energy Sources — Energy sources that can be replenished naturally in a short time. Examples: solar, wind, hydroelectric, tidal, geothermal, biomass.

Non-Renewable Energy Sources — Energy sources that are finite and cannot be replenished quickly. Examples: coal, oil, natural gas, nuclear fuels (uranium).

Fossil Fuels — Coal, oil, and natural gas formed from the remains of prehistoric organisms. They are non-renewable and produce CO₂ when burned.

Solar Energy — Energy from sunlight. Used via photovoltaic cells (direct electricity) or solar thermal collectors (heating).

Wind Energy — Kinetic energy of wind captured by wind turbines to generate electricity.

Hydroelectric Energy — Energy from flowing or falling water used to spin turbines and generate electricity.

Nuclear Energy — Energy released from splitting (fission) or combining (fusion) atomic nuclei.

[illegible]

MCQ's

1. Work is defined as:

A. Force $\times$ velocity B. Force $\times$ displacement C. Force $\times$ time D. Mass $\times$ acceleration

Answer: B

2. SI unit of work is:

A. Newton B. Watt C. Joule D. Pascal

- **Kinetic energy** is the energy a body possesses due to its motion. $KE = \frac{1}{2}mv^2$.

• **Potential energy** is the energy stored in a body due to its position or state. Gravitational PE = mgh.

Q3. State the law of conservation of energy with an example.

Ans: The law of conservation of energy states that energy cannot be created or destroyed; it can only be transformed from one form to another. The total energy of an isolated system remains constant.

Example: In a swinging pendulum, potential energy converts to kinetic energy and back. At the highest point, energy is all potential; at the lowest point, it is all kinetic. Total mechanical energy remains constant (ignoring friction).

Q4. Define power. Write its SI unit.

Ans: Power is the rate at which work is done or energy is transferred. It is given by $P = W / t$, where W is work and t is time. The SI unit of power is the watt (W). $1 \text{ W} = 1 \text{ J/s}$.

Q5. What is meant by efficiency? Why is it always less than 100%?

Ans: Efficiency is the ratio of useful output energy (or power) to the total input energy (or power), expressed as a percentage. It is always less than 100% because some energy is always lost to the surroundings, usually as heat due to friction, air resistance, or electrical resistance.

Q6. Differentiate between renewable and non-renewable energy sources. Give one example of each.

Ans:

• **Renewable energy sources** can be replenished naturally in a short time. Example: solar energy.

• **Non-renewable energy sources** are finite and take millions of years to form. Example: coal.

Q7. What is the difference between positive and negative work? Give examples.

Ans:

• **Positive work** occurs when force and displacement are in the same direction. Example: a person pushing a box forward.

• **Negative work** occurs when force and displacement are in opposite directions. Example: friction acting on a sliding box (friction opposes motion, so work by friction is negative).

Q8. How is electrical energy generated from hydroelectric power plants?

Ans: In a hydroelectric power plant, water stored at a height has gravitational potential energy. When released, this water flows downward, converting PE to kinetic energy. The fast-flowing water strikes turbine blades, causing them to spin. The turbine is connected to a generator, which converts mechanical energy into electrical energy.

Q9. What is a fossil fuel? Why are they considered non-renewable?

Ans: Fossil fuels (coal, oil, natural gas) are formed from the remains of dead plants and animals buried under the Earth's surface for millions of years under high pressure and temperature. They are considered non-renewable because their formation takes millions of years, and they are being consumed much faster than they are formed.

Q10. A body is thrown vertically upward. Describe the energy changes taking place.

Ans: When a body is thrown vertically upward:

- At the start (ground level): Maximum kinetic energy, minimum (zero) potential energy.
- As it rises: KE decreases (velocity decreases), PE increases (height increases).
- At the highest point: Minimum (zero) KE, maximum PE.
- During descent: PE converts back to KE.

At every point, the sum of KE and PE remains constant (ignoring air resistance).

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Long Questions

Q1. Explain the concept of work in physics. Discuss the three cases of work with examples.

Ans: In physics, work is defined as the product of force and displacement in the direction of the force. Mathematically, $W = F \times d \times \cos \theta$, where θ is the angle between force and displacement vectors.

Case 1: Positive Work ($\theta = 0^\circ$)

When force and displacement are in the same direction, maximum work is done.

$W = Fd$ (since $\cos 0^\circ = 1$)

Example: Pushing a trolley forward. The force applied and displacement are in the same direction.

Case 2: Negative Work ($\therefore \vec{F} \cdot \vec{d} < 0$)

When force and displacement are in opposite directions, work is negative.

$$W = -Fd \text{ (since } \cos 180^\circ = -1 \text{)}$$

Example: Braking force applied to a moving car. The brakes apply force opposite to the direction of motion.

Case 3: Zero Work ($\therefore \vec{F} \cdot \vec{d} = 0$)

When force is perpendicular to displacement, no work is done.

$$W = 0 \text{ (since } \cos 90^\circ = 0 \text{)}$$

Example: A person carrying a suitcase while walking on a level road. The force (upward to hold the bag) is perpendicular to the horizontal displacement. Another example is when displacement is zero (pushing a wall).

Q2. Derive the formula for kinetic energy. Explain how kinetic energy depends on mass and velocity.

Ans: Kinetic energy is the energy possessed by a body due to its motion.

Derivation:

Consider a body of mass m initially at rest ($v_i = 0$). A force F acts on it, causing displacement d and accelerating it to velocity v .

$$\text{Work done: } W = F \times d$$

$$\text{From Newton's second law: } F = ma$$

$$\text{From equations of motion: } v^2 = v_i^2 + 2ad \Rightarrow v^2 = 0 + 2ad \Rightarrow d = \frac{v^2}{2a}$$

$$\text{Substituting: } W = ma \times \left(\frac{v^2}{2a}\right) = \frac{1}{2}mv^2$$

This work done on the body is stored as kinetic energy.

$$KE = \frac{1}{2}mv^2$$

Dependence:

- KE is directly proportional to mass (double mass $\Rightarrow$ double KE).
- KE is proportional to the square of velocity (double velocity $\Rightarrow$ quadruple KE).

Example: A truck moving at 20 m/s has more KE than a car of lower mass moving at the same speed. A car moving at 40 m/s has four times the KE of the same car moving at 20 m/s.

Q3. Explain the law of conservation of energy. Give three real-life examples of energy transformations.

Ans: The law of conservation of energy states that energy cannot be created nor destroyed; it can only be transformed from one form to another. The total energy of an isolated system always remains constant.

Examples of Energy Transformations:

Electric Fan: Electrical energy $\Rightarrow$ Kinetic energy of blades (mechanical energy) + Heat energy (due to resistance in motor windings)

Solar Cell: Light energy (from sun) $\Rightarrow$ Electrical energy

Car Engine: Chemical energy (from fuel) $\Rightarrow$ Heat energy (combustion) $\Rightarrow$ Kinetic energy (motion of pistons) $\Rightarrow$ Mechanical energy (wheels rotate)

In every transformation, the total energy before equals the total energy after. Some energy is always converted to heat (due to friction and resistance), which dissipates to the surroundings but is not destroyed.

Q4. What is power? Derive the relationship $P = Fv$. How is power related to the rate of doing work?

Ans: Power is defined as the rate at which work is done or energy is transferred.

$$\text{Mathematically: } P = \frac{W}{t}$$

Derivation of $P = Fv$:

$$\text{Work done: } W = F \times d$$

$$\text{Power: } P = \frac{W}{t} = \frac{(F \times d)}{t}$$

$$\text{Since } \frac{d}{t} = v \text{ (velocity): } P = F \times v$$

This means that power is the product of force and average velocity. For a constant force, more power is required to move an object at a higher speed.

Units:

- SI unit: watt (W), $1 \text{ W} = 1 \text{ J/s}$
- Practical unit: kilowatt (kW) = 1000 W
- Electrical energy unit: kilowatt-hour (kWh), $1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$

Q5. Compare renewable and non-renewable energy sources. Discuss the advantages and disadvantages of solar, wind, and nuclear energy.

Availability	Unlimited (replenished naturally)	Limited (finite reserves)
Environmental Impact	Low (clean energy)	High (pollution, greenhouse gases)
Cost	High initial, low running	Lower initial, high running
Examples	Solar, wind, hydro	Coal, oil, gas, nuclear

- **Advantages:** Abundant, free, clean, no pollution, low maintenance
- **Disadvantages:** Intermittent (night/cloudy), high initial cost, requires large area, energy storage needed

- **Advantages:** Clean, renewable, low operating cost, good in coastal/hilly areas
- **Disadvantages:** Unpredictable (wind speed varies), noisy, visual pollution, requires large land area, kills birds

- **Advantages:** Produces huge energy from small fuel, no greenhouse gases during operation, reliable base-load power
- **Disadvantages:** Radioactive waste disposal problem, risk of accidents (Chernobyl, Fukushima), high decommissioning cost, non-renewable (uranium is finite), potential for weaponization

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Q1. Calculate the work done when a force of 25 N moves a box through 4 m in the direction of force.

$$\begin{aligned} W &= F \times d \\ W &= 25 \times 4 \\ W &= 100 \text{ J} \end{aligned}$$
[illegible]

Force required to lift = Weight = $mg = 15 \times 10 = 150 \text{ N}$
 Work done = $F \times d = 150 \times 2 = 300 \text{ J}$
 (Alternatively: $W = mgh = 15 \times 10 \times 2 = 300 \text{ J}$)

[illegible]
$$\begin{aligned} KE &= \frac{1}{2}mv^2 = \frac{1}{2} \times 1000 \times (20)^2 \\ KE &= 500 \times 400 \\ KE &= 200,000 \text{ J} = 200 \text{ kJ} \end{aligned}$$

KE , = $\frac{1}{2} \times 1000 \times (40)^2 = 500 \times 1600 = 800,000 \text{ J} = 800 \text{ kJ}$
KE becomes 4 times (since KE $\propto v^2$).

[illegible]

Q4. A stone of mass 5 kg is dropped from a height of 10 m. Find its potential energy at the top and

$$P = 600 \text{ W}$$